ELECTRICAL SURGE INCIDENT RESPONSE

Hardware Failure Diagnosis, Component Isolation & Micro-ATX Rebuild

Analyst: Bryan Sykes | August 2026 | Hardware Incident Portfolio Project

1. Incident Overview

Project Title	Electrical Surge Incident Response & Full PC Rebuild
Date	August 2026
Analyst	Bryan Sykes
Trigger Event	Thunderstorm-related electrical surge; wall outlet and surge protector failure
System Affected	Desktop PC on AM5 platform, Ryzen 5 9600X, RTX 4080 Super
Root Cause	ROG Strix B850-A Gaming WiFi motherboard is the sole failed component
Diagnostic Method	Direct visual inspection + component substitution testing (no assumptions)
Classification	CONFIDENTIAL - Personal Hardware Incident, Portfolio Documentation

During an overnight thunderstorm, a power flash caused an entire room circuit to lose power while the PC was being powered on. Rather than assuming the surge protector alone absorbed the event, the wall outlet and internal PC components were inspected directly and tested systematically. This report documents the full incident response: electrical hazard assessment, component-by-component fault isolation, root cause confirmation, and the subsequent full micro-ATX system rebuild, including a factory hardware defect discovered during reassembly.

2. Electrical Hazard Assessment

Severity: HIGH	**Area:** Wall Outlet / Room Circuit	**Evidence:** Figure 1

What Happened

A power flash caused total room power loss while the PC was being powered on. Rather than assuming the surge protector alone absorbed the event, the wall outlet was inspected directly before any component was re-powered.

Findings

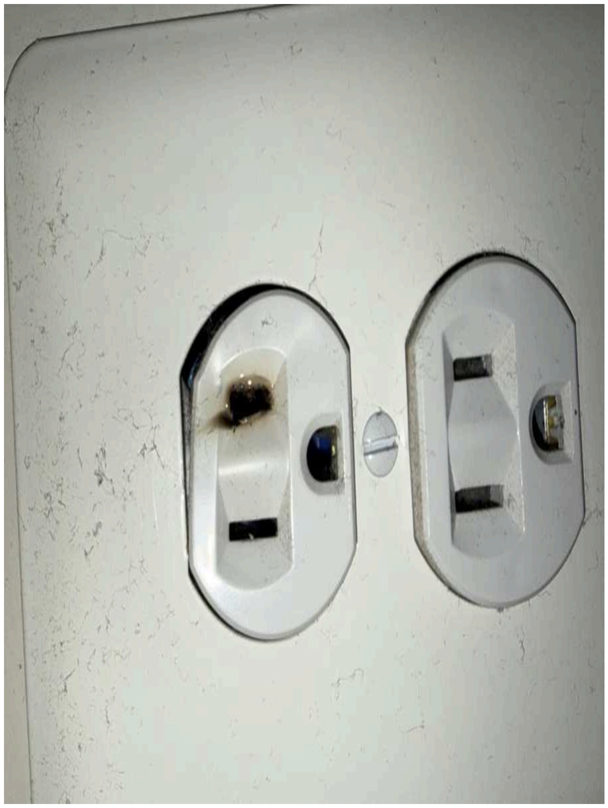

Fig. 1 — *Wall outlet inspected after the surge, visible scorching/burn damage on the hot-side contact, confirming the failure was not isolated to the surge protector.*

Remediation

- Breaker for the affected room was located and shut off before further inspection or component handling.
- The damaged surge protector (Lvetek, scorched prong) was retired.
- The wall outlet was replaced before any component was reconnected to that circuit.

3. Component Isolation Testing

With the system failing to POST after the surge, each major component was isolated through direct substitution rather than assumption. Testing one variable at a time and documenting the result before moving to the next.

3.1 Power Supply Unit

Test Method: Direct Substitution	**Result:** Cleared	**Evidence:** Figure 2

Diagnostic Steps

- The original DeepCool 1000W PSU delivered standby power (fans/LEDs active) but the system still failed to POST.
- A second, brand-new MSI PSU was substituted as a control test. The identical failure persisted, clearing the PSU as the root cause.

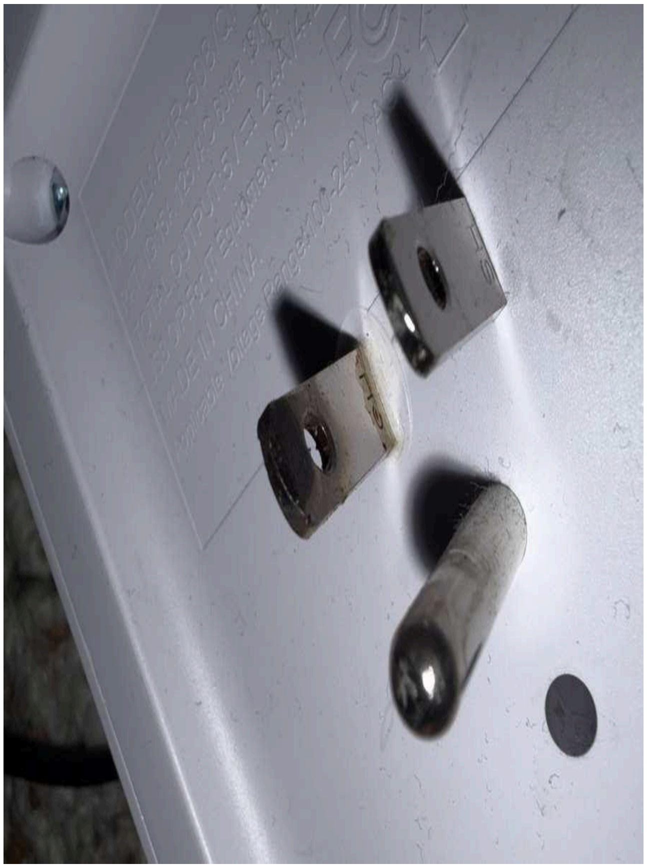

Fig. 2 — *Fried damage from the wall surge protector*

Outcome

- PSU cleared as a factor. The control unit was returned for a full refund.

3.2 Graphics Card

Test Method: Direct Substitution (2nd GPU)	**Result:** Cleared	**Evidence:** Figures 3–5

Diagnostic Steps

- Primary GPU (PNY XLR8 RTX 4080 Super) was reseated and its 12VHPWR-style power connector inspected for pin damage. No discoloration or melting found.

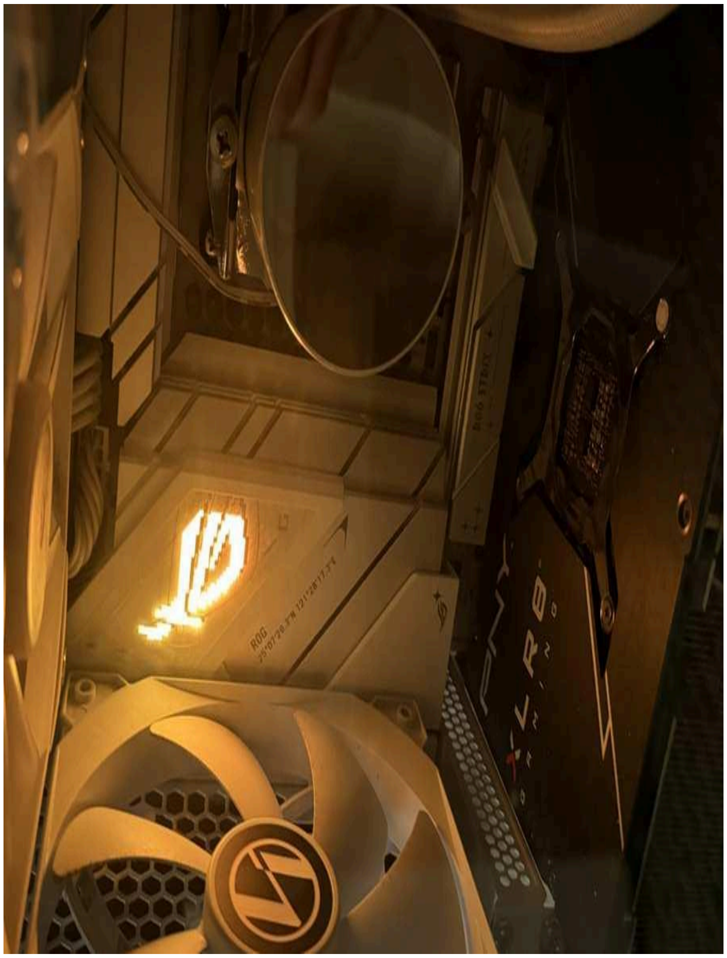

Fig. 3 — *PNY RTX 4080 Super installed with the motherboard's illuminated diagnostic LED visible this VGA-stage indicator was the primary symptom tracked throughout testing. Light shows that DeepCool PSU is still supplying motherboard with idle standby power*

- A second, known-good GPU (Sparkle-branded card, sourced from a separate working system) was substituted into the same PCIe slot as a control test.

Fig. 4 — *Known-good Sparkle Intel Arc B570 GPU installed in the same PCIe slot. The identical diagnostic LED failure occurred, ruling out both graphics cards as the cause.*

Fig. 5 — *Sparkle GPU seated alongside the M.2 heatsink shroud, confirming slot fit and full seating during the control test.*

Outcome

- Both graphics cards cleared as a factor.

3.3 Memory

Test Method: Visual Inspection + Barebones Boot	**Result:** Cleared	**Evidence:** Figures 6–7

Diagnostic Steps

- T-Force Delta DDR5 RGB memory (2x16GB kit) was removed and the gold contacts visually inspected for corrosion, bending, or discoloration.

Fig. 6 — *Single memory module inspected at the contact edge. No damage or discoloration present.*

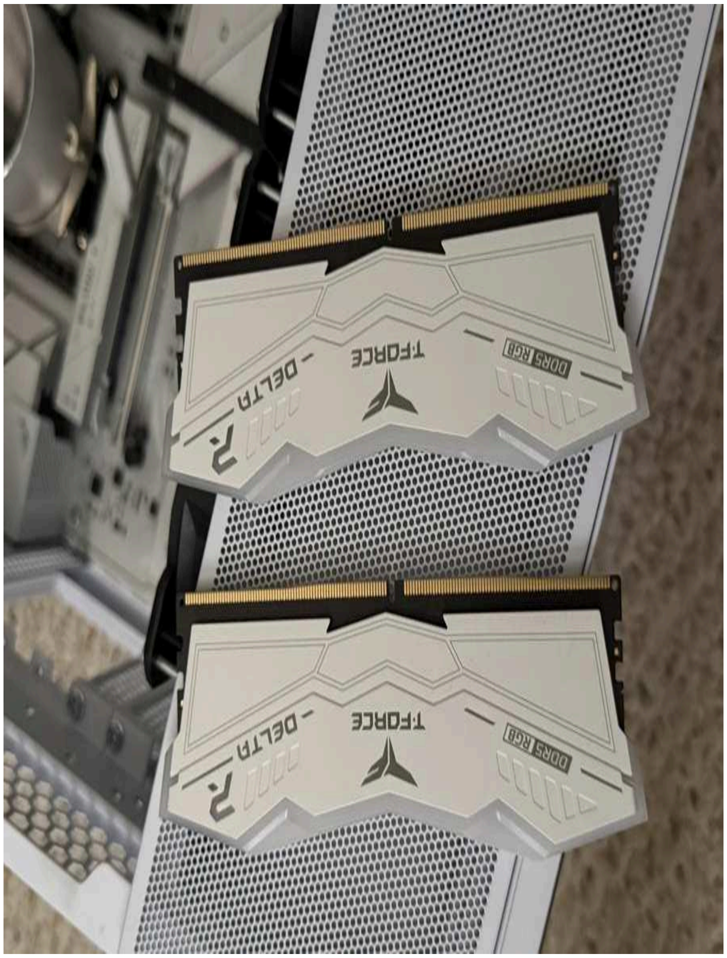

Fig. 7 — *Full memory kit laid contact-side out for side-by-side comparison. Both modules were confirmed to be physically undamaged.*

- A barebones boot test was performed using a single RAM stick with no GPU and no drives installed. The system still failed at the same diagnostic stage.

Outcome

- The memory kit cleared as a factor.

3.4 Storage

Test Method: External Enclosure Verification	**Result:** Cleared — 4/4 Healthy	**Evidence:** N/A (verified on separate laptop)

Diagnostic Steps

- All 4 SSDs were removed from the system and individually verified using an external USB enclosure connected to a separate laptop.

Outcome

- All 4 drives confirmed readable with no data loss. Storage cleared, all user data protected independent of the desktop's condition.

3.5 CPU and Socket

Test Method: Visual Inspection + Full Barebones POST	**Result:** Inconclusive — narrowed to CPU/Motherboard	**Evidence:** Figures 8–9

Diagnostic Steps

With PSU, GPU (x2), RAM, and storage all cleared via direct substitution, remaining suspects were narrowed to the CPU and motherboard.

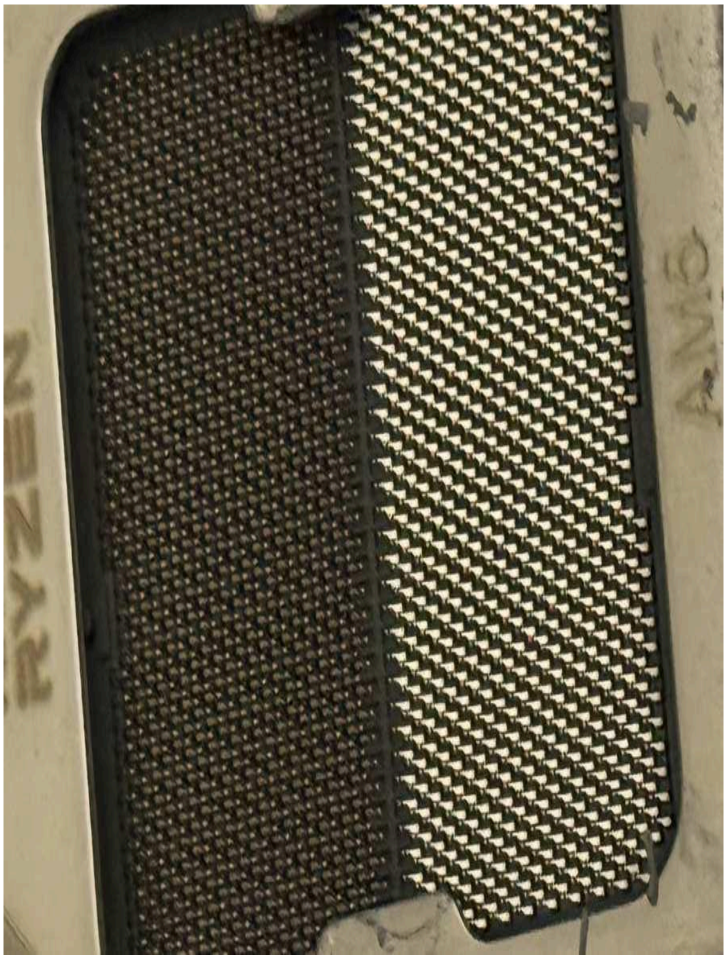

Fig. 8 — *AMD Ryzen 5 9600X CPU contact pads inspected directly for scorching, pitting, or corrosion — surface shows normal wear consistent with routine reseating, no damage found.*

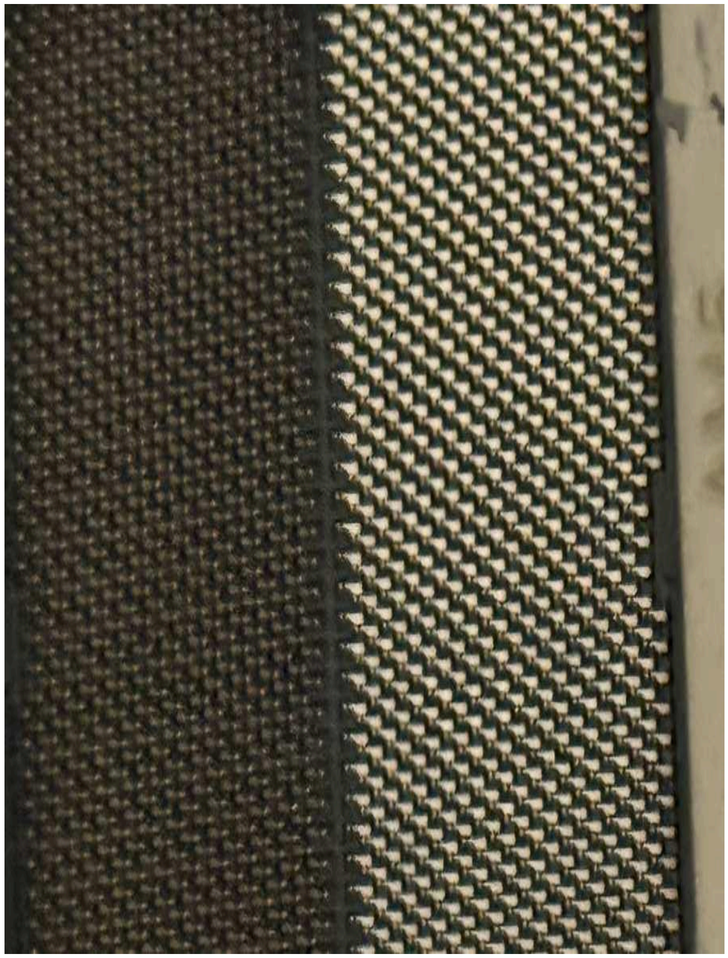

Fig. 9 — *CPU contact pads re-photographed under different lighting to rule out damage obscured by glare or angle — confirmed clean.*

- A full barebones test (CPU installed, single RAM stick, no GPU, no drives) still failed to reach POST, narrowing the fault to the CPU or the motherboard itself.

Outcome

- Without a spare AM5 motherboard or CPU available to cross-test the two remaining suspects independently, a replacement motherboard became the only remaining path to isolate the fault.

4. Root Cause Confirmation

Component: ASUS ROG Strix B850-A Gaming WiFi	**Result:** CONFIRMED FAULT	**Evidence:** Figures 10–11

Diagnostic Steps

A new ASUS ROG Strix B850-A Gaming WiFi motherboard was sourced and installed using the original CPU, RAM, and GPU.

Fig. 10 — *ROG Strix B850-A Gaming WIFI Motherboard determined to have a fried PCIe slot and fried VGA capabilities, rendering the system unable to successfully go through the Power-On-Self-Test sequence (POST).*

Result

The system POSTed and booted successfully on first power-on with the replacement board. This confirmed the motherboard as the sole component damaged by the electrical surge. Every other tested component (PSU, both GPUs, RAM, and all 4 SSDs) was verified fully functional, avoiding unnecessary replacement spending on parts that did not need it.

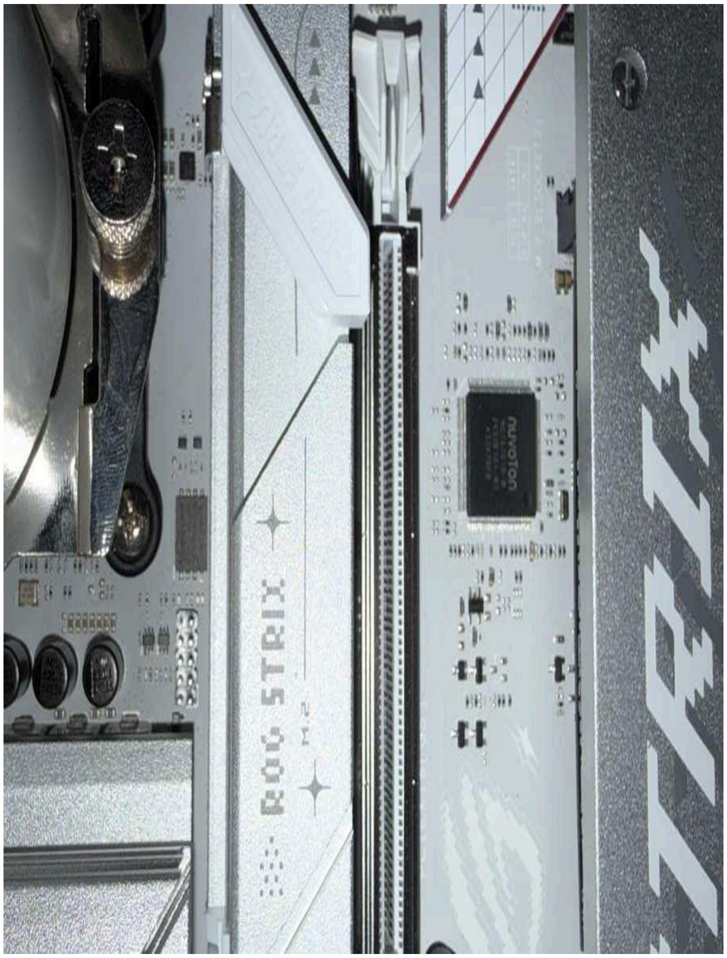

Fig. 11 — *Onboard I/O controller (Nuvoton) and M.2 heatsink region of the replacement board inspected post-install as part of final verification.*

5. Full Rebuild — Micro-ATX Downsize

The surge event was used as an opportunity to downsize the build into a Lian Li A3-mATX case, requiring case-fit verification (GPU length, PSU length, cooler clearance) prior to purchase and full hardware reassembly.

5.1 Factory Hardware Defect — Stripped Screws

Severity: MEDIUM	**Scope:** 6 screws — PSU bracket, radiator mount, SSD mount	**Evidence:** Figures 12–15

What Happened

During reassembly, screw heads across multiple mounting points were found to be pre-stripped from the factory, rather than damaged during installation.

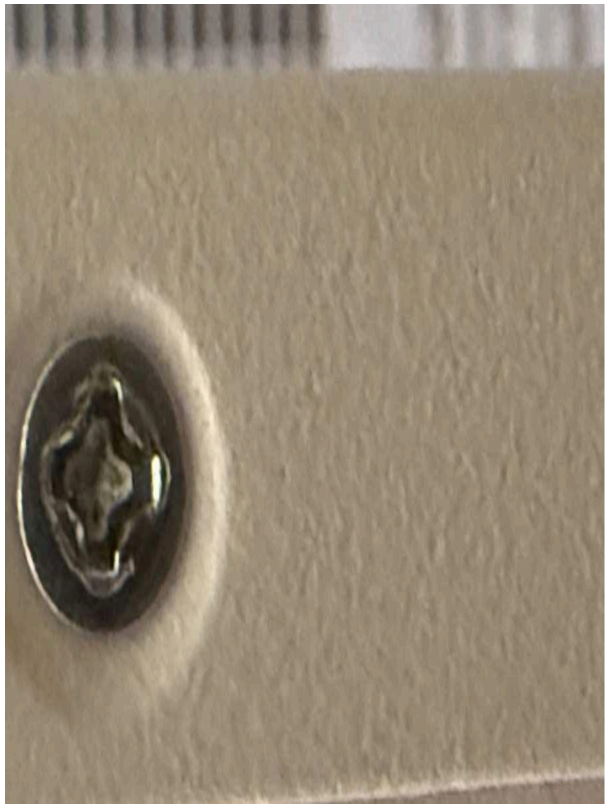

Fig. 12 — *First stripped screw discovered — PSU mounting bracket. Cross-slot pattern shows visible rounding consistent with over-torqued factory assembly.*

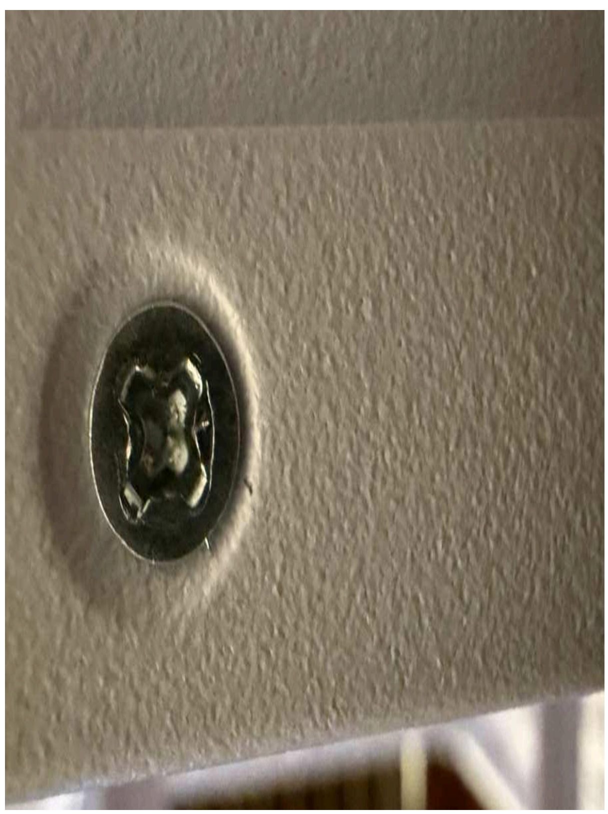

Fig. 13 — *Second PSU bracket screw — more significant stripping, cross pattern nearly smoothed over.*

Fig. 14 — *Front I/O panel screws inspected for comparison — cross-slot patterns intact and undamaged, confirming the stripping defect was isolated to specific mounting points, not case-wide.*

Fig. 15 — *All 6 affected screws laid out together after removal — each was successfully extracted using a rubber-band-and-driver friction technique.*

Root Cause

- Diagnosed as factory over-torquing during assembly, not user error confirmed by comparing against undamaged front-panel screws from the same case.

Resolution

- All 6 screws were removed via rubber-band-and-driver friction technique.
- Correct replacement thread spec (6-32 pan-head) identified before sourcing new screws, after an initial mismatch with a countersunk-head screw from a spare parts bag was caught and corrected.

5.2 System Reassembly & Verification

Check: fTPM/PSP NV Reset	**Risk:** Data lockout if BitLocker enabled	**Result:** Safely Accepted — No Data Risk

- On first boot with the new motherboard, an fTPM/PSP NV reset prompt appeared — expected behavior following a CPU/motherboard re-pairing.
- BitLocker had been proactively disabled on the boot drive prior to the swap, so the reset was accepted with no data-loss risk.
- Final build completed with full cable management, RGB memory installed, and a 240mm AIO liquid cooler with an integrated LCD pump display.

6. Troubleshooting Log

Issue	Root Cause	Resolution
Total room power loss on PC power-on	Electrical surge during thunderstorm damaged wall outlet and surge protector simultaneously	Breaker shut off immediately; outlet visually inspected and replaced before any component was re-powered
System failed to POST after surge	Motherboard sustained surge damage (confirmed only after full component isolation)	Systematically substituted PSU, GPU, RAM, and storage; replaced motherboard once all other components were cleared
GPU appeared to seat but pulled out easily	Anti-sag GPU support bracket was pushing the card up and out of full contact with the PCIe slot	Removed the support bracket, reseated the GPU fully into the Q-Release Slim latch, confirmed firm seating
PSU bracket screws would not back out	Screws were pre-stripped from the factory (confirmed via comparison to undamaged front-panel screws)	Removed using rubber-band-and-driver friction technique; sourced correct 6-32 pan-head replacements
Replacement screw from spare parts bag did not match	Bag was labeled/stocked with countersunk-head screws, not the pan-head type originally used	Visually compared head profile against the original screw before installation instead of trusting the bag label alone
fTPM/PSP NV reset prompt on first boot with new board	Expected behavior when a CPU is paired with a new motherboard	Confirmed BitLocker was already disabled on the boot drive, then accepted the fTPM reset with no data risk

7. Key Technical Lessons

- A tripped or dead surge protector does not confirm the full extent of surge damage. The outlet and wiring behind it must be independently inspected before any component is re-energized.
- Component isolation should proceed by direct substitution wherever possible (a second known-good GPU, a second PSU) rather than by inference. It is the only way to conclusively clear a component.
- A full barebones boot test (CPU + single RAM stick, no GPU, no drives) is the most efficient way to narrow a no-POST fault down to the CPU/motherboard pairing once peripherals are cleared.
- Anti-sag GPU brackets, if adjusted incorrectly, can prevent a graphics card from fully seating in its PCIe slot, a mechanical issue that can visually mimic a hardware failure.
- Factory-stripped fasteners are a real and recurring defect category, not just user error. Comparing suspect screws against known-undamaged screws from the same product is a fast, reliable way to confirm the defect.
- Print/packaging labels on spare hardware (e.g., screw bags) should be visually verified against the actual part in hand labels that can be mismatched or misleading.
- Disabling BitLocker before a planned CPU/motherboard change removes the single biggest risk (data lockout) associated with an fTPM reset.

8. Conclusion

This incident was resolved through a disciplined, evidence-based troubleshooting process: verifying the electrical hazard directly rather than assuming the surge protector absorbed it, isolating each hardware component through direct substitution rather than guesswork, and confirming root cause before any replacement purchase. This approach identified the motherboard as the single failed component while the PSU, both tested GPUs, the RAM kit, and all four SSDs were verified fully functional and preserved without unnecessary spending.

The subsequent rebuild uncovered and resolved a secondary factory hardware defect — six pre-stripped screws — using the same evidence-first methodology: comparing suspect parts against known-good controls before drawing conclusions. This dual outcome, a confirmed electrical root cause and a confirmed manufacturing defect, demonstrates the same diagnostic discipline applied throughout this project's home lab and security work: verify before you conclude, isolate before you replace.

Documentation prepared by: Bryan Sykes | August 2026 | Hardware Incident Portfolio Project